



Upgrading ethanol to *n*-butanol over highly dispersed Ni–MgAlO catalysts



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ABSTRACT

Ethanol is widely used as a fuel blender for gasoline. However, it has a low energy content and blending ratio. The fuel quality can be improved by upgrading ethanol to high-energy-content alcohols such as *n*-butanol. Here, we synthesized high-carbon alcohols from ethanol over Ni-based catalysts in a fixed-bed reactor. The selectivities for *n*-butanol and C₄–C₈ alcohols reached 55.2% and 85%, respectively, at 18.7% ethanol conversion over the optimal catalyst, Ni₄MgAlO, with an areal reaction rate of 62.6 μmol h^{−1} m^{−2} at 523 K. The effects of Ni promotion and acid–base site distributions of catalysts were investigated using various techniques. The results show that metallic Ni sites promoted ethanol dehydrogenation, and the released hydrogen spilled over the catalyst, leading to in situ hydrogenation of reaction intermediates. Strong basic and moderately acidic sites were needed for aldol condensation reactions. A reaction network was proposed based on product analysis under different conditions.

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1. Introduction

Ethanol is the major renewable chemical derived from biomass, and is mainly produced via fermentation of hexose sugars derived from crops [1]. In the past five years, the fermentation of biomass, especially lignocellulosic biomass, to ethanol has been intensively investigated and rapid progress has been made [2,3]. It is anticipated that the global production of ethanol will surpass 30 billion gallons in 2017 [4]. Ethanol is a versatile feedstock that can be used as a fuel additive or solvent. It can also be catalytically transformed to important chemicals such as ethylene, *n*-butanol, acetaldehyde [5], isobutene [6–8], and *p*-xylene [9,10].

Ethanol is primarily used as a green fuel additive to alleviate the problems of fossil fuel depletion and environmental pollution. In Brazil, gasoline is blended with more than 10 v/v% ethanol, taking advantage of abundant sugarcane resources for fermentation and local policy incentives [11,12]. The use of ethanol-blended gasoline is also politically encouraged in many countries, including the United States and China. Nevertheless, the ethanol content of gasoline is limited to below 20% to avoid affecting the fuel quality and damaging internal combustion engines [13]. The blending ratio can be increased and the fuel quality improved by upgrading ethanol to high-carbon alcohols, typically *n*-butanol, which have higher

energy content, blending ratios, and research octane numbers and properties similar to those of gasoline [14].

Catalytic conversion of ethanol to *n*-butanol involves the Guerbet reaction, where a primary aliphatic alcohol is converted into its β-alkylated dimer alcohol with loss of one equivalent of water [15]. In 1931, ethanol coupling was first observed on mixtures of MgO, Al₂O₃, and CuO_x [16]. Three types of heterogeneous catalysts have been developed for this reaction: metal oxides, hydroxyapatite (HAP), and transition-metal catalysts. Metal oxides are promising catalysts for ethanol conversion because they give high product selectivities and controllable synthetic processes; for instance, Mg–Al oxide catalysts with different structures show high selectivities for *n*-butanol in ethanol conversion. HAP is a thermally stable catalyst with a tunable number of acid and base sites, the distributions of which can be changed by varying the preparation conditions [17–19]. Metal oxides and HAP catalysts both have acid and base sites, which are active in aldol condensation reactions [20–22]. However, these catalysts are relatively inert in ethanol activation, and elevated temperatures (higher than 573 K) are needed for ethanol conversion. Hence, various transition metals have been exploited to enhance ethanol activation, including Cu, Ni, Co, Fe, Pd, Rh, Ru, and Au, some of which have shown high activity in ethanol conversion under relatively mild conditions [23–30].

The coupling of ethanol to give *n*-butanol involves several consecutive steps: ethanol dehydrogenation, aldol addition, dehydration, and hydrogenation reactions. Catalysts should have

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balanced numbers of different types of active sites, including metal, acid, and base sites. Layered double hydroxides (LDHs) such as hydrotalcite are promising precursors for controllable synthesis of catalysts with multifunctional sites [31]. Highly dispersed metals can be introduced, and the distribution of acid and base sites can be tuned to some extent; therefore LDHs can be used to clarify the key factors affecting the catalytic conversion of ethanol to *n*-butanol.

In this study, a series of highly dispersed Ni–MgAlO catalysts were synthesized from LDH precursors and used for ethanol conversion. The acid and base sites of Ni–MgAlO catalysts were tuned to minimize side reactions at high ethanol conversions. The effects of the Mg/Al ratio, Ni content, and reaction temperature on the product distribution were investigated. The reasons for the high performance of the Ni4MgAlO catalyst were identified by comparing it with NiAlO and Ni4MgO catalysts in a variety of characterizations. A reaction network was proposed based on analysis of the reaction products under different conditions.

2. Experimental

2.1. Catalyst preparation

Oxides of Mg, Al, and Ni–MgAlO catalysts were prepared from LDH precursors using a coprecipitation method, as described elsewhere [32,33]. Taking the Ni4MgAlO catalyst as an example, an aqueous solution (50 mL) containing $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (0.02 mol), $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.02 mol), and $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (0.08 mol) was added dropwise to a continuously stirred solution (50 mL) containing Na_2CO_3 and NaOH in a molar ratio of 2:3. The pH of the obtained suspension was adjusted to 10 by 3 mol L^{-1} NaOH aqueous solution. The suspension was kept at 338 K for 18 h and a precipitate with a hydrotalcite structure was formed. The hydrotalcite was separated by filtration, repeatedly washed with deionized water, and dried at 353 K. Finally, the precursor was calcined at 873 K in air for 2 h and then reduced at 873 K in pure hydrogen for 1 h. The prepared catalyst was denoted by Ni4MgAlO, indicating Ni:Mg:Al molar ratios of 1:4:1.

Ni/MgAlO catalysts were prepared using a deposition–precipitation method. Typically, an aqueous solution (5 mL) containing Ni $(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (0.49 g) and an aqueous solution of NaOH (0.2 mol L^{-1}) were simultaneously pumped into a mixture of an MgAlO support (2 g) and water (10 mL) at a rate of 1 mL min^{-1} . The solution pH was adjusted to 10 using 3 mol L^{-1} NaOH. The suspension was filtered off and reduced at 873 K in a hydrogen atmosphere.

2.2. Catalyst characterization

X-ray diffraction (XRD) patterns were recorded with a PANalytical X'Pert-Pro powder X-ray diffractometer, using $\text{CuK}\alpha$ monochromated radiation ($k = 0.1541 \text{ nm}$) at a scanning speed of $5^\circ/\text{min}$ over the range $10\text{--}80^\circ$.

N_2 adsorption and desorption isotherms were measured at 77 K using a Micromeritics ASAP 2460 system. The Brunauer–Emmett–Teller method was used to calculate the catalyst surface area.

X-ray photoelectron spectroscopy (XPS) was performed with an ESCALAB 250Xi spectrometer using an $\text{AlK}\alpha$ ($h\nu = 1486.6 \text{ eV}$) X-ray source with a pass energy of 20 eV and an analysis chamber base pressure less than $1 \times 10^{-8} \text{ Pa}$. Before the experiments, the samples were pressed into thin discs and prereduced with hydrogen at 873 K for 1 h and then passivated using 1% O_2/He for 1 h at ambient temperature. Ar sputtering to a depth of 3 nm was performed before data collection because of catalyst surface oxidation.

The Ni content of the catalysts was determined using inductively coupled plasma atomic emission spectrometry (Thermo IRIS Intrepid II).

Transmission electron microscopy (TEM), high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM), and energy-dispersive X-ray spectroscopy (EDS) elemental mapping were performed using a JEM-2100F microscope, operated at a voltage of 200 kV. Before analysis, the sample was dispersed in ethanol and then mounted on C-coated Cu grids, followed by slow evaporation of the solvent at room temperature.

Temperature-programmed desorption (TPD) experiments were performed using a Micromeritics AutoChem II 2920 automated system. Samples for H_2 -TPD tests were first reduced in situ under hydrogen at 873 K for 1 h. The freshly reduced catalysts were then flushed with Ar for 60 min at a temperature 10 K higher than that for reduction to remove the extra hydrogen adsorbed on the surface. Afterward, the temperature was decreased to 523 K and pure H_2 was injected until saturation was reached. Finally, after the sample was flushed with Ar until the baseline was stable, TPD was performed from 523 to 800 K at a rate of 10 K min^{-1} .

H_2 and CO_2 microcalorimetric measurements were performed at 313 K using a BT 2.15 heat-flux calorimeter. The calorimeter was connected to a gas-handling and volumetric system with MKS 698A Baratron capacitance manometers for precision pressure measurements ($\pm 1.33 \times 10^{-2} \text{ Pa}$). The maximum apparent leak rate of the volumetric system was $10^{-4} \text{ Pa min}^{-1}$, with a system volume of approximately 80 cm^3 . The ultimate dynamic vacuum of the system was ca. 10^{-5} Pa . NH_3 microcalorimetric measurements were performed at 423 K with a system volume of approximately 60 cm^3 .

Fourier transform infrared spectroscopy (FT-IR) was performed in the transmission mode, using a BRUKER Equinox 55 spectrometer, at a resolution of 4 cm^{-1} in the range $4000\text{--}400 \text{ cm}^{-1}$. An IR cell interfaced with a custom-built glass vacuum manifold was used for sample pretreatment and gaseous titrants. Prior to characterization, the prepared catalyst wafer was pretreated under hydrogen at 673 K for 60 min. The wafer was cooled to 303 K under a dynamic vacuum. The background spectrum was collected and then pyridine (Alfa Aesar, >99%) or CO_2 (>99.9%) adsorption was performed at 303 K for 3 min and the spectrum was recorded. Finally, the wafer was heated to a specified temperature and then the temperature was decreased to 303 K in a dynamic vacuum to monitor the disappearance of IR peaks from bound titrant species.

2.3. Catalytic reactions

All experiments were performed in a high-pressure automated continuous-flow fixed-bed reactor (Scheme S1 in the Supplementary Material). In a typical experiment, the catalyst (1.5 g) was placed at the center of a steel reactor with 50–70 mesh quartz on both sides of the reactor. After in situ reduction of the catalyst at 873 K with pure hydrogen at 60 mL min^{-1} for 1 h, the reactor was cooled to the reaction temperature. Then the gas was switched to N_2 at 30 mL min^{-1} , and analytically pure ethanol (99.5 wt.%) was pumped into the reactor using a high-performance liquid chromatography pump. The liquid products were collected in a gas–liquid separator. The gaseous products were analyzed using an Agilent gas chromatography (GC) system equipped with a TDX-01 column. The liquid products were analyzed using an Agilent GC system equipped with a Varian CP-WAX58 (FFAP) CB capillary column.

The ethanol conversion products were qualitatively analyzed using GC–mass spectrometry (MS, Varian). The injector temperature was 523 K, with a split ratio of 10:1. The MS operating conditions were as follows: the ion source temperature, interface temperature, and quadrupole temperature were set at 503, 523,

and 313 K, respectively. The electron ionization mode was set at 70 eV with a mass range of $m/z = 29$ –450.

The product yields, ethanol conversion, selectivity, and reaction rate were calculated as

$$Y (\%) = \frac{\text{Moles of carbon in the product}}{\text{Moles of carbon in ethanol fed to the reactor}} \times 100,$$

$$\text{Con.} (\%) = \frac{\text{Moles of ethanol reacted}}{\text{Moles of ethanol fed to the reactor}} \times 100,$$

$$\text{Sel.} (\%) = \frac{\text{Moles of carbon in the target product}}{\text{Moles of carbon in ethanol reacted}} \times 100,$$

Areal reaction rate ($\text{mol h}^{-1} \text{m}^{-2}$)

$$= (\text{ethanol conversion}/100)/[\text{surface area of catalyst} \\ \times \text{catalyst weight (g)}/\text{ethanol flow rate (mol h}^{-1})],$$

where the moles of ethanol reacted were calculated based on the moles of carbon in all liquid and gaseous products.

3. Results and discussion

3.1. Catalytic conversion of ethanol

3.1.1. Ethanol conversion over various catalysts

The Mg/Al ratio in oxide catalysts greatly affects the *n*-butanol selectivity and ethanol conversion [34]; therefore the Mg/Al ratio in the Ni/MgAlO catalysts was first optimized. As shown in Fig. 1, the ethanol conversion over the 5%Ni/Al₂O₃ catalyst was 25%, with 35% selectivity for *n*-butanol. Large amounts of diethyl ether and ethylene were produced; this is attributed to dehydration reactions occurring at strong acid sites on the catalyst [6,35]. To lower the density of strong acid sites, MgO was introduced into the carrier, using a coprecipitation method. The ethanol conversion decreased with increasing Mg/Al ratio, but the selectivity for high-carbon alcohols increased. At an Mg/Al ratio of 4:1, the selectivity for *n*-butanol increased to 50.2%, and the selectivity for C₄–C₈ alcohols reached 77.3%. The Mg/Al ratio was therefore set at 4:1 for catalyst evaluation to obtain high yields of primary alcohols.

The effects of the Ni content on alcohol selectivity and ethanol conversion were investigated. As shown in Fig. 2, over the mixed oxide catalyst in the absence of Ni, the predominant product was *n*-butanol, with a selectivity of 58%, but the ethanol conversion was 0.5%. Increasing the Ni content from zero to 19.5 wt.% led to a monotonic increase in ethanol conversion from 0.5% to 18.8%,

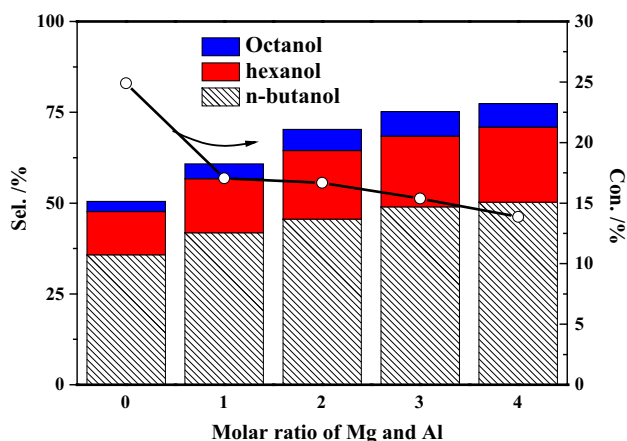


Fig. 1. Ethanol conversions over 5%Ni/MgAlO catalysts with various Mg/Al ratios. Conditions: N₂ 3 MPa, 30 mL/min; 99.5% ethanol; 523 K; WHSV 3.2 h⁻¹.

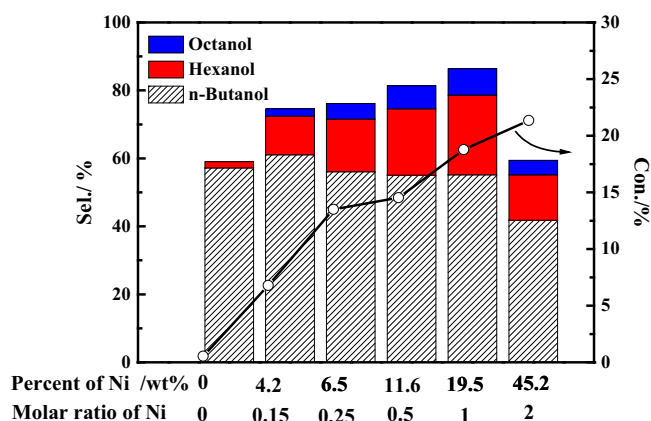


Fig. 2. Ethanol conversions over Ni-MgAlO catalysts with various Ni content. Conditions: N₂ 3 MPa, 30 mL/min; 99.5% ethanol; 523 K; WHSV 3.2 h⁻¹.

without a decrease in the selectivity for *n*-butanol. Over the optimal Ni4MgAlO (19.5 wt.% Ni) catalyst, the selectivities for *n*-butanol and total C₄–C₈ alcohols reached 55.2% and 85%, respectively (ratios of C₄:C₆:C₈ were 8:3:1). However, further increasing the Ni content to 45.2 wt.% significantly decreased the alcohol selectivity. A large amount of gaseous byproducts was produced, including hydrogen, carbon monoxide, methane, and carbon dioxide; these products accounted for ca. 15% ethanol conversion (based on carbon).

The dehydrogenation of ethanol to acetaldehyde is regarded as the rate-determining step in ethanol conversion under mild conditions [24,36]. When metallic Ni sites were introduced into the catalysts, the ethanol conversion improved significantly, with considerable *n*-butanol selectivity. On the other hand, the presence of Ni also promoted carbon–carbon bond cleavage in ethanol, particularly when the amount of Ni was excessive. This is consistent with previous reports that Ni particles promoted ethanol cracking reactions, following the route $\text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_4 + \text{H}_2 + \text{CO}$ [37,38]. The optimal molar ratio of Ni for ethanol conversion was therefore 1; i.e., Ni4MgAlO was the optimal catalyst and was used for further study. On the other hand, ethanol coupling is a cascade reaction. The produced *n*-butanol can be further dehydrogenated to butanal, which can react with itself or acetaldehyde to form C₆, C₈, and higher-carbon products through aldol condensation. After hydrogenation reactions, hexanol, octanol, and high-carbon alcohols were produced [36,39].

The Ni4MgAlO catalyst shows significant advantages over those catalysts reported in the literature for ethanol conversion to *n*-butanol. Unlike most oxides, HAP, or metal-enhanced catalysts, reaction conditions for the Ni4MgAlO catalyst were mild and ethanol conversion was relatively high (reaction temperature: 523 K vs. >573 K; ethanol conversion ca. 20% vs. <10%) [29,34,40]. The weight hourly space velocity (WHSV) of ethanol conversion over the Ni4MgAlO catalyst was also higher than that for most metal catalysts, with a higher alcohol selectivity [25,28].

3.1.2. Effects of reaction temperature on alcohol selectivity

The product distribution in ethanol conversion is sensitive to the reaction temperature [4]. The effects of temperature on ethanol conversion and alcohol selectivity were therefore investigated. As shown in Fig. 3, when the reaction temperature was increased from 473 to 548 K, the ethanol conversion increased linearly from 7.2% to 22.1%, and the selectivity for C₄–C₈ alcohols remained above 80% over the Ni4MgAlO catalyst. The improved ethanol conversion indicates that ethanol is relatively inert under mild conditions, but is activated at high temperature. This is consistent with previous studies, in which reaction temperatures as high as 673 K were used

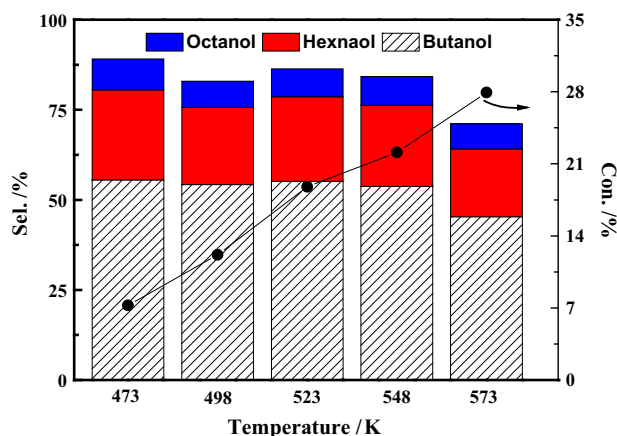


Fig. 3. Ethanol conversions over the Ni4MgAlO catalyst at various temperatures. Conditions: N₂ 3 MPa, 30 mL/min; 99.5% ethanol; WHSV 3.2 h⁻¹.

for ethanol conversion over composite oxide catalysts [34,40]. Interestingly, the selectivity for alcohols at 473–548 K was very stable, giving a wide operating-temperature window. Further increasing the temperature to 573 K increased the ethanol conversion to 27.9%, but the overall selectivity for C₄–C₈ alcohols decreased to 70%. Based on the reaction efficiency and alcohol selectivity, the reaction temperature was kept below 548 K.

3.1.3. Stability of optimal catalyst

The stability of the Ni4MgAlO catalyst was tested at 523 K at a WHSV of 4 h⁻¹. As shown in Fig. 4, the initial selectivities for *n*-butanol, hexanol, and octanol were 58.3, 15.3, and 3.8%, respectively, at 11.0% ethanol conversion. After reaction for 60 h, the ethanol conversion and alcohol selectivity were unchanged, showing that the catalyst had good stability under the present reaction conditions. The used catalyst was characterized using TEM (see Supplementary Material, Figs. S3C–S3D). Ni particles in the used catalyst are highly dispersed with average particle size of 5.6 nm, which is similar to that of the fresh catalyst. The catalyst showed high stability under reaction conditions.

3.1.4. Reactivity of comparison catalysts

The reasons for the high performance of the Ni4MgAlO catalyst were investigated by synthesizing NiAlO, Ni4MgO, Ni3AlO, and

Ni6MgO catalysts and comparing their ethanol conversions with that for Ni4MgAlO. The reaction rate was normalized based on the surface areas of the catalysts because multiple active sites were involved in the reaction.

The reaction activities and selectivities for ethanol conversions over typical comparison catalysts are shown in Table 1. Over NiAlO (entry 2), a catalyst with the same Ni:Al molar ratio as the Ni4MgAlO catalyst, the areal reaction rate reached 235.2 μmol h⁻¹ m⁻² at 523 K. When the Ni content was decreased to 18.7%, i.e., the Ni3AlO catalyst, which has a Ni percentage similar to that of Ni4MgAlO, an areal reaction rate of 74.4 μmol h⁻¹ m⁻² was obtained (entry 3). The order of the activities over the three catalysts was NiAlO > Ni3AlO > Ni4MgAlO. These results indicate that both metallic Ni and Al oxide promote ethanol conversion. However, the selectivities for *n*-butanol over the NiAlO and Ni3AlO catalysts were inferior to that over Ni4MgAlO. Although the areal reaction rates were controlled at a low level, 19–24 μmol h⁻¹ m⁻², with ca. 7% ethanol conversion, the selectivity was still less than 36% over the first two catalysts, lower than that over the Ni4MgAlO catalyst (55.5%). Large amounts of gaseous products were detected; this is attributed to the decomposition of products over abundant Ni sites and the dehydration of ethanol over acid sites on these two catalysts [41].

For the Ni–MgO catalysts of Ni4MgO and Ni6MgO (entries 4 and 5), which have Ni:Mg molar ratio and Ni content similar to those for the Ni4MgAlO catalyst, the areal reaction rates at 523 K were 40.0 and 38.0 μmol h⁻¹ m⁻², respectively, notably lower than that over the Ni4MgAlO catalyst (62.6 μmol h⁻¹ m⁻², entry 1). Evidently, high MgO content in Ni–MgO catalysts inhibit ethanol conversion. The selectivity for *n*-butanol over the Ni4MgAlO catalyst was much higher than over the Ni–MgO catalysts (55% vs. 46%). At a similar ethanol conversion of ca. 7%, the areal reaction rate over Ni6MgO was 99.2 μmol h⁻¹ m⁻², much higher than that over the Ni4MgAlO catalyst, but the *n*-butanol selectivity decreased to 43.0% because of the side reactions induced by the higher reaction temperature (543 K).

A comparison of the results obtained using different catalysts shows that both metallic Ni and alumina in catalysts promoted ethanol conversion. The presence of Mg was important for *n*-butanol selectivity. Over the Ni4MgAlO catalyst, the amounts of metallic Ni, acid, and base sites is optimal, resulting in relatively high *n*-butanol selectivity and ethanol conversion.

The structures and surface adsorption properties of Ni4MgAlO, NiAlO, and Ni4MgO catalysts were investigated and compared to clarify the reasons for the high activity of the Ni4MgAlO catalyst, as discussed in the next section.

3.2. Catalyst characterization

3.2.1. XRD and STEM

Hydrotalcites are anionic clays consisting of brucite-like layers and are ideal precursors for preparing highly dispersed catalysts. As shown in Fig. 5, major diffraction peaks at 42.9° and 62.3° in the XRD patterns of the Ni–MgAlO catalysts can be indexed to the (200) and (220) planes of MgO (PDF #00-004-0829); this is attributed to the abundant MgO in hydrotalcites. Ni peaks were not detected until the molar content of Ni reached 1 (19.5 wt.%); this indicates that the Ni particles were highly dispersed in the composite oxides. When the molar ratio of Ni was increased to 2, peaks from metallic Ni (PDF # 01-070-0989) were more clearly observed. The average crystal sizes of metallic Ni particles in the Ni4MgAlO and 2Ni4MgAlO catalysts were estimated to be 5.3 and 5.8 nm, respectively, using the Scherrer equation. This shows that increasing the Ni content of the hydrotalcites did not induce Ni particles aggregation, and the Ni particles were highly dispersed throughout the mixed metal oxides.

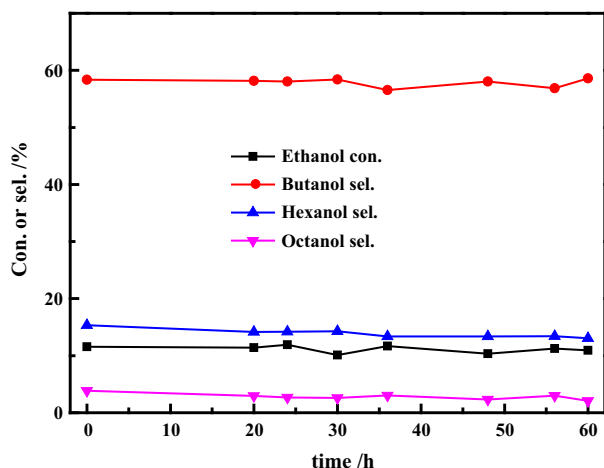
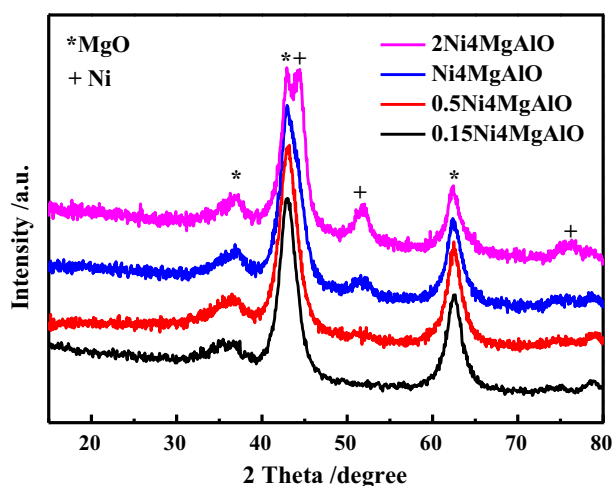


Fig. 4. Stability of the Ni4MgAlO catalyst. Conditions: N₂ 3 MPa, 30 mL/min; 99.5% ethanol; 523 K; WHSV 4 h⁻¹.

Table 1Comparison of ethanol conversion over various catalysts under the conditions N₂ 3 MPa, 30 mL/min, 99.5% ethanol.

Entry	Catalyst	Ni content (wt.%)	Areal reaction rate ($\mu\text{mol h}^{-1} \text{m}^{-2}$) ^a	Reaction temperature (K)	Con. (%)	Selectivity (%)		
						<i>n</i> -Butanol	C ₆	C ₈
1	Ni4MgAlO	19.5	23.9	473	7.2	55.5	25.0	8.6
			62.6	523	18.8	55.2	23.4	7.7
2	NiAlO	54.0	24.2	468	7.4	30.2	4.9	–
			235.2	523	72.0	11.1	2.1	–
3	Ni3AlO	18.7	19.4	473	7.0	35.1	0.5	–
			74.4	523	27.1	39.7	8.4	1.5
4	Ni4MgO	22.4	40.0	523	5.6	45.7	2.0	–
			50.0	533	7.0	42.9	2.3	–
5	Ni6MgO	20.1	38.0	523	4.6	46.3	1.8	–
			99.2	543	7.5	43.0	1.3	–

^a Surface area and pore distribution of catalysts are measured and presented in the [Supplementary Material, Fig. S1](#).**Fig. 5.** XRD patterns of Ni–MgAlO catalysts.

The XRD patterns of NiAlO, Ni4MgO, and the hydrotalcite precursor of Ni4MgAlO were also obtained for comparison (see the [Supplementary Material, Fig. S2](#)). The average crystal size of metallic Ni particles in the NiAlO catalyst was ca. 5 nm, even when the Ni content was 54%. The size of the Ni particles in the Ni4MgO catalyst was similar to that in the NiAlO catalyst. These results indicate that the Ni particle size was not sensitive to the support and Ni content when the catalysts were prepared from LDH precursors. The Ni particle size therefore did not affect ethanol conversion.

TEM, HAADF-STEM, and EDS images were obtained to clarify the morphology and element distribution of the Ni4MgAlO catalyst. As shown in [Fig. 6A and B](#), Ni particles of size 4–6 nm were uniformly distributed, and the lattice distance of 0.203 nm was associated with metallic Ni, which is consistent with the XRD results. The HAADF images and maps of Ni, Mg, and Al for the Ni4MgAlO catalyst show that Ni, Mg, and Al were homogeneously distributed, which is further evidence of high dispersion of Ni in the Mg and Al oxides. A large-scale image of the Ni–MgAlO catalyst (see the [Supplementary Material, Fig. S3A](#)) shows that the Ni4MgAlO catalyst retained the original lamellar structure of the hydrotalcite precursor; this also confirms that Ni particles were highly dispersed via the LDH precursors prepared using the coprecipitation method.

3.2.2. Hydrogen adsorption and desorption properties of Ni4MgAlO and comparison catalysts

Ni particles promoted ethanol conversion and gave considerable alcohol selectivity. Ethanol dehydrogenation and hydrogenation

reactions involve hydrogen adsorption and desorption on catalysts, and this process needs to be investigated [4,21]. We studied the hydrogen adsorption properties of typical catalysts using microcalorimetry. [Fig. 7](#) (left) shows the differential heats of H₂ adsorption at 313 K as a function of H₂ uptake over the NiAlO, Ni4MgO, and Ni4MgAlO catalysts. The amount of H₂ adsorbed at saturation on the NiAlO catalyst was 497.8 $\mu\text{mol g}^{-1}$, which is higher than those on the Ni4MgO (102.4 $\mu\text{mol g}^{-1}$) and Ni4MgAlO (163.8 $\mu\text{mol g}^{-1}$) catalysts. The corresponding initial heats of H₂ adsorption were 99, 83, and 103 kJ mol^{−1}, respectively, indicating that the H₂ adsorption strengths were similar on the Ni4MgAlO and NiAlO catalysts. The percentages of active sites in the heat distribution histograms ([Fig. 7](#) (right)) show that 62.4% of the active sites on the NiAlO catalyst had a differential heat of H₂ adsorption greater than 80 kJ mol^{−1}; 87.6% of the active sites on the Ni4MgO catalyst had a differential heat of H₂ adsorption less than 80 kJ mol^{−1}. For the Ni4MgAlO catalyst, the differential heats of H₂ adsorption were much more uniformly distributed, with 42.9% of active sites in the range 80–100 kJ mol^{−1}. These results indicate that the Ni4MgAlO catalyst had moderate-strength hydrogen adsorption at 313 K, which may facilitate hydrogen evolution during the reaction.

In addition to hydrogen adsorption, hydrogen transfer for hydrogenation of the butanol precursor, crotonaldehyde, could also be crucial for the formation of saturated alcohols. The desorption of hydrogen from various catalysts was investigated using an H₂-TPD experiment at the reaction temperature. Before desorption, the catalysts were reduced in a hydrogen flow at 873 K and then flushed with Ar at 883 K for 1 h to remove surface-adsorbed hydrogen. Hydrogen adsorption was performed at 523 K, which is the reaction temperature for ethanol conversion. TPD was performed after saturation adsorption and flushing with Ar to achieve a stable baseline of the thermal conductivity detector signal. As shown in [Fig. 8](#), the amount of H₂ released by the Ni4MgAlO catalyst was significantly larger than the amounts released by the NiAlO and NiMgO catalysts after saturated adsorption of hydrogen at 523 K. This indicates that ethanol-released hydrogen atoms could be stored or transferred over the surface of the Ni4MgAlO catalyst, which could be further used for in situ hydrogenation of unsaturated products. In contrast, the amounts of hydrogen adsorbed on the NiAlO and Ni4MgO catalysts were much smaller, making the catalyst less effective in the hydrogen-borrowing reaction [42].

The microcalorimetric adsorption and H₂-TPD results show that the Ni4MgAlO catalyst had a moderate adsorption capacity but gave the largest amount of desorbed hydrogen. These contradictory results could be caused by the different conditions used for these experiments. Microcalorimetric adsorption experiments were conducted at 313 K under vacuum. The amount of hydrogen

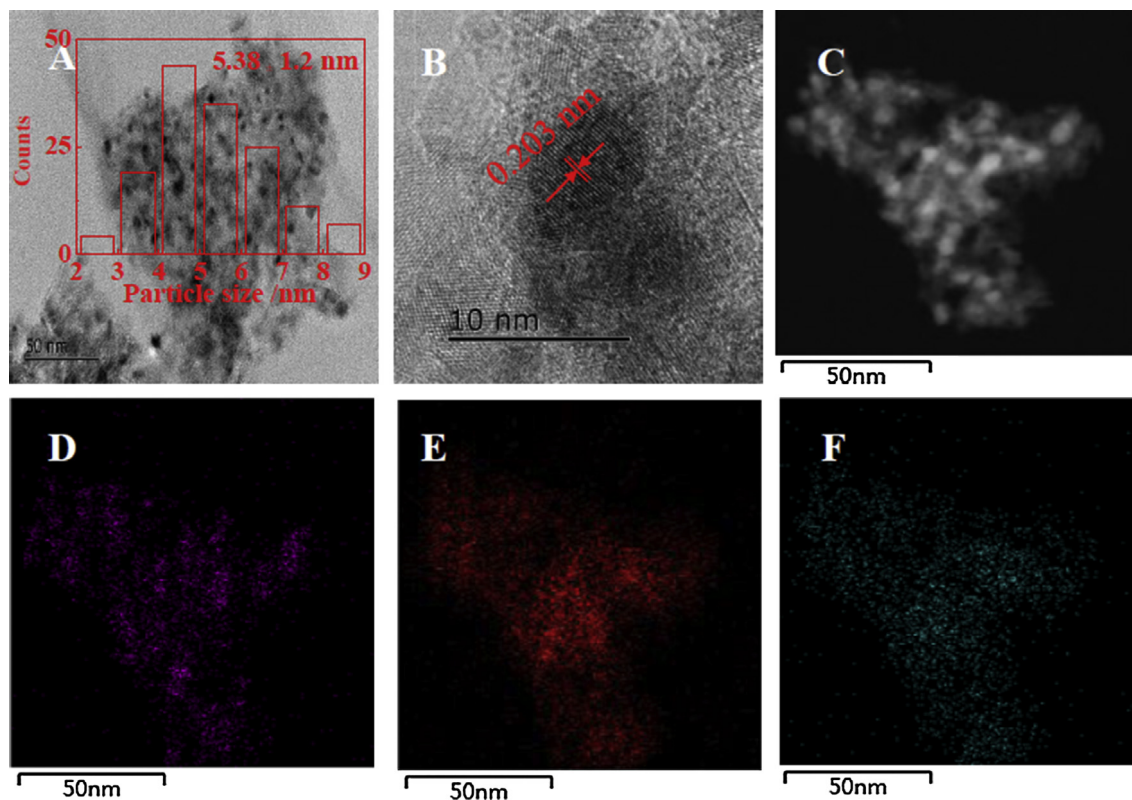


Fig. 6. TEM images (A, B), HAADF-STEM images (C), and EDS elemental mappings of Ni (D), Mg (E), and Al (F) for the Ni₄MgAlO catalyst.

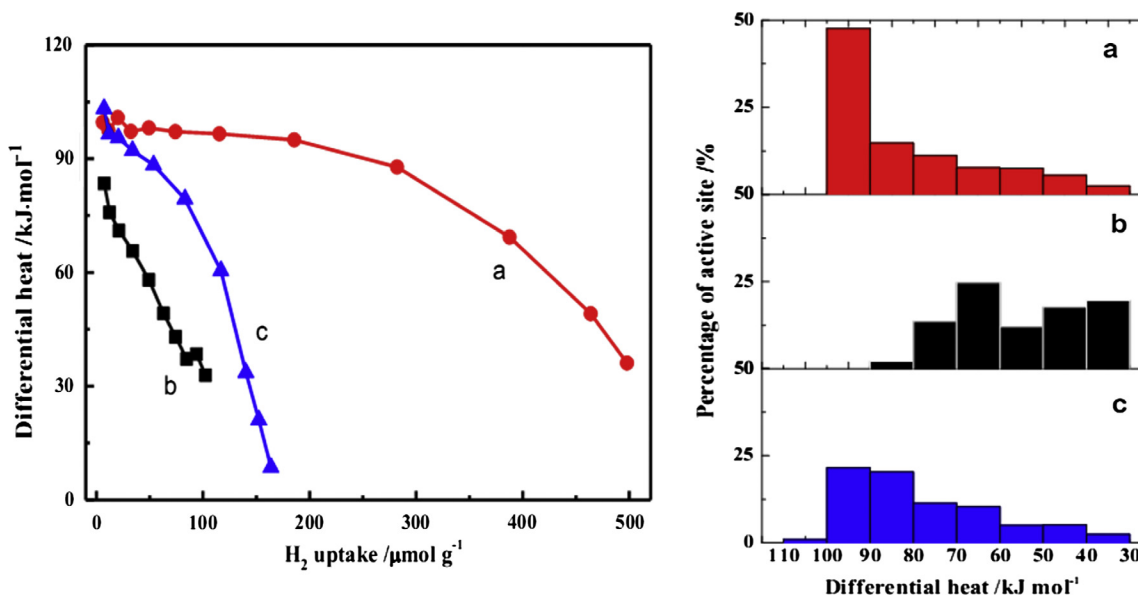


Fig. 7. Differential heats of H₂ adsorption on (a) NiAlO catalyst, (b) Ni₄MgO catalyst, and (c) Ni₄MgAlO catalyst at 313 K in a static system (left) and corresponding heat distribution histograms (right).

adsorbed should be closely related to the number of metallic Ni sites on the catalyst. The XPS results (see the [Supplementary Material, Fig. S4](#)) show that the percentages of metallic Ni in NiAlO, Ni₄MgAlO, and Ni₄MgO were 41.5, 17.1, and 2.6%, respectively; this is consistent with the microcalorimetric adsorption result. However, the H₂-TPD experiments were conducted at 523 K. At this temperature, hydrogen spillover was promoted, and more

hydrogen atoms were trapped on defects in the Mg-rich mixed oxide catalysts; therefore the Ni₄MgAlO catalyst gave the highest desorption [34,43].

In the conversion of ethanol to *n*-butanol, ethanol is dehydrogenated and produces mobile hydrogen atoms, which are stored or transferred over the catalyst for further hydrogenation reactions [43,44]. Metallic Ni in the Ni₄MgAlO catalyst greatly promoted

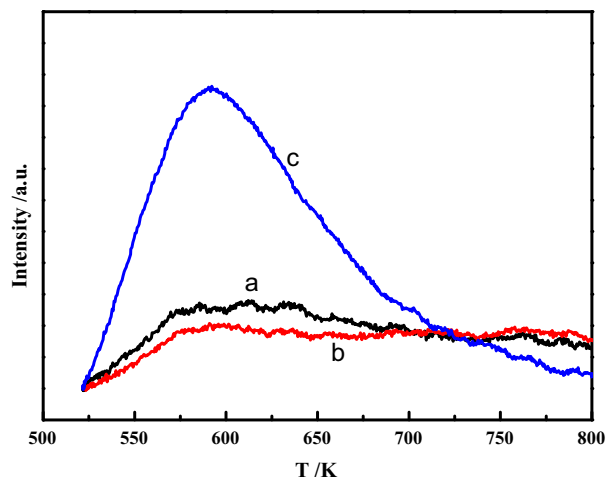


Fig. 8. H_2 -TPD for (a) NiAlO catalyst, (b) Ni4MgO catalyst, and (c) Ni4MgAlO catalyst.

ethanol dehydrogenation, and hydrogen atoms were stored and transferred for the hydrogenation of intermediates to produce primary alcohols.

3.2.3. Microcalorimetric adsorption of CO_2 and NH_3 on Ni4MgAlO and comparison catalysts

During ethanol conversion, aldol addition and dehydration of the aldol addition products are determined by the acid and base sites of the catalyst. The acidic and basic properties of typical catalysts were therefore investigated using microcalorimetry.

Fig. 9 shows the CO_2 adsorption heats for NiAlO, Ni4MgO, and Ni4MgAlO as a function of CO_2 coverage. The initial heat of CO_2 adsorption decreased in the order Ni4MgO (118 kJ mol^{-1}) > Ni4MgAlO (117 kJ mol^{-1}) > NiAlO (108 kJ mol^{-1}). However, the Ni4MgAlO catalyst had a maximum adsorption amount, $352 \mu\text{mol g}^{-1}$. A better view of the base sites' energy distribution is shown in Fig. 9 (right). More than 50% of the adsorption heat on the Ni4MgO catalyst is distributed in the range $100\text{--}130 \text{ kJ mol}^{-1}$; this is attributed to strong base sites [44]. In contrast, the main adsorption heat

for the NiAlO catalyst is associated with moderate base sites, and ranges from 80 to 110 kJ mol^{-1} . For the Ni4MgAlO catalyst, both strong and moderate base sites were detected in the heat range $80\text{--}120 \text{ kJ mol}^{-1}$; this is consistent with the FT-IR spectra for CO_2 adsorption (see the [Supplementary Material, Fig. S5](#)). On a basic material such as MgO, the coordinately unsaturated O^{2-} at the edges and corners of crystallites are usually considered to be strong base sites with an adsorption heat higher than 130 kJ mol^{-1} [45]. The introduction of Ni slightly decreases the basicity of MgO; this can be attributed to coverage by Ni and strong interaction between Ni and MgO. Similarly, the moderate base sites in NiAlO are associated with the hydroxyl groups [46], which are also changed by the introduction of metallic Ni. Over the Ni4MgAlO catalyst, the coordinately unsaturated O^{2-} and hydroxyl groups were both uniformly distributed and of moderate strength.

The acidic properties of the catalysts were determined microcalorimetrically, using NH_3 as a probe molecule. As shown in Fig. 10, the initial adsorption heats for the NiAlO and Ni4MgAlO catalysts were 104.3 and 88.5 kJ mol^{-1} , respectively, with similar adsorption amounts. In contrast, there was only physical adsorption of NH_3 on the Ni4MgO catalyst. The heat distribution histograms in Fig. 10 (right) show that there were strong and medium-strong acid sites on the NiAlO catalyst, with adsorption heats between 80 and 110 kJ mol^{-1} . In contrast, no acid sites were detected on the Ni4MgO catalyst. For the Ni4MgAlO catalyst, all the adsorption heats were below 90 kJ mol^{-1} ; this is attributed to medium-strong and weak acid sites on the catalyst. The acidic properties of Ni4MgAlO catalysts were also determined from FT-IR spectra of pyridine adsorption (see the [Supplementary Material, Fig. S6](#)). Only Lewis acid sites were observed (peak centered at 1450 cm^{-1}). The intensity of the peaks in the IR spectra decreased greatly after thermal treatment at 423 K , suggesting that most of the Lewis acid sites on the Ni4MgAlO catalyst are weak. This is consistent with microcalorimetric results. The surfaces of composite MgO and Al_2O_3 oxides are enriched with aluminum oxide, as previously reported [20], and this may be why the NH_3 adsorption properties of the NiAlO and Ni4MgAlO catalysts are similar. Meanwhile, MgO can cover the strong acid sites of alumina, and this reduces the number of strong acid sites on the Ni4MgAlO catalyst. The addition of Ni also covered some acid sites, which decreased the adsorption heat even for the NiAlO catalyst [46,47].

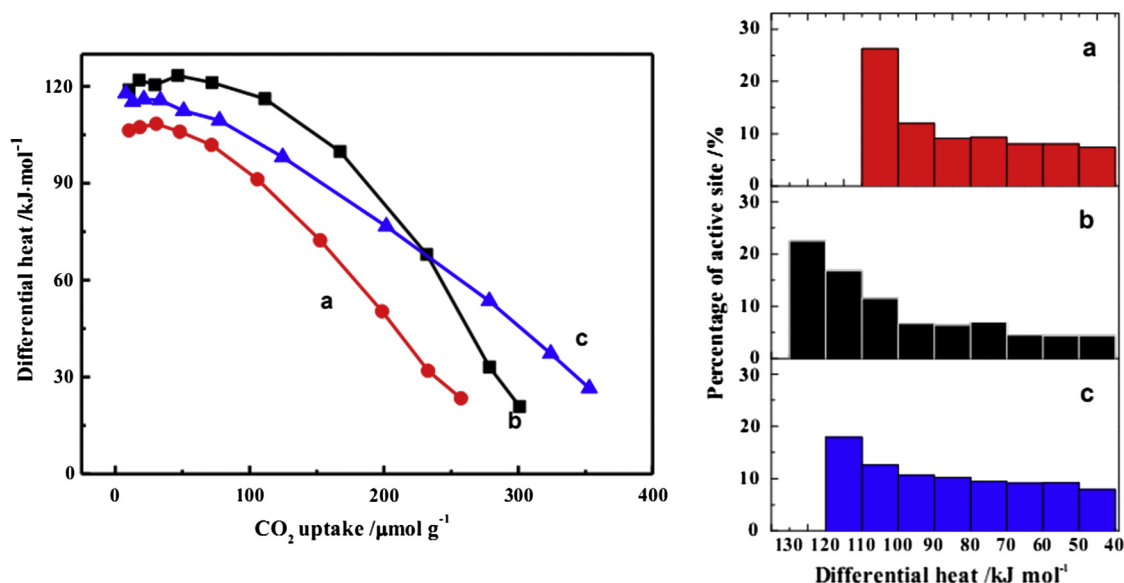


Fig. 9. Differential heats of CO_2 adsorption on (a) NiAlO catalyst, (b) Ni4MgO catalyst, and (c) Ni4MgAlO catalyst at 313 K in a static system (left) and corresponding heat distribution histograms (right).

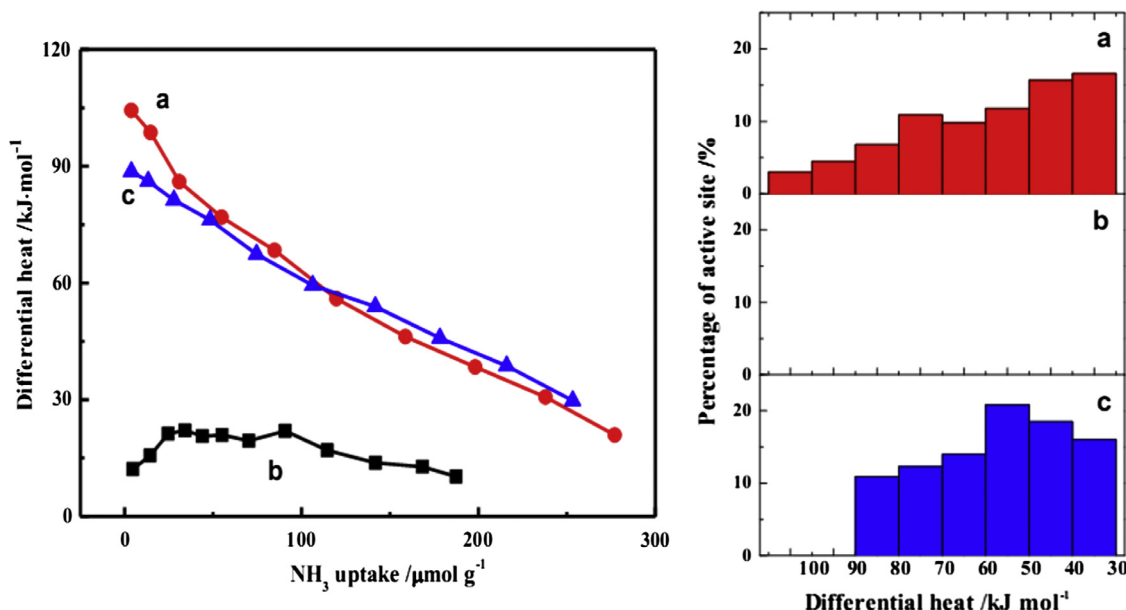


Fig. 10. Differential heats for NH₃ adsorption on (a) NiAlO catalyst, (b) Ni4MgO catalyst, and (c) Ni4MgAlO catalyst at 423 K in a static system (left) and corresponding heat distribution histograms (right).

The metal, acid, and base site distributions of typical catalysts were summarized (Table S1 in the Supplementary Material) based on the results for H₂, CO₂, and NH₃ microcalorimetric adsorption. The active sites of catalysts were classified into three types according to the adsorption heat: strong sites with adsorption heats greater than 100 kJ mol⁻¹, medium sites with adsorption heats between 80 and 100 kJ mol⁻¹, and weak sites with adsorption heats between 30 and 80 kJ mol⁻¹ [48]. Medium metal sites, medium acid sites, and strong base sites on catalysts were regarded as key factors for *n*-butanol production, and their number ratios were calculated. The NiAlO catalyst had a medium metal, medium acid, and strong base sites ratio of 9.5:1:2.1. In contrast, the Ni4MgO catalyst had weak metal sites, strong base sites, and no acid sites. Over the Ni4MgAlO catalyst, the number ratio of medium metal, medium acid, and strong base sites is 2.4:1:3.8. For ethanol conversion, metal sites on the catalysts enhance the hydrogen-transfer reaction [49], and acid sites of the catalysts promote ethanol activation. The ethanol conversion over the NiAlO catalyst reached a high value of 72.0%, resulting from promotion by these two types of sites. In contrast, only 5.6% ethanol was converted on the Ni4MgO catalyst; this should be ascribed to the low metal and acid site content and the high content of base sites on the catalyst. Strong acid sites on the catalyst promote dehydrogenation and dehydration reactions, but induce side reactions, e.g., inter/intra dehydration of ethanol. In contrast, a catalyst only containing strong base sites will be less active in ethanol dehydration and disadvantaged for condensation reactions to some extent. The ratio of metal, acid, and base sites should be balanced to obtain a high *n*-butanol selectivity. On the Ni4MgAlO catalyst, medium metal, medium acid, and strong base sites were balanced to be 2.4:1:3.8, which promoted ethanol conversion without depressing acetaldehyde condensation, and resulted in relatively high alcohol yields.

3.3. Reaction network for ethanol conversion over Ni–MgAlO catalysts

The reaction network for ethanol conversion was investigated by GC–MS analysis of the products over typical catalysts at different WHSV (see the Supplementary Material, Fig. S7). The main product over the Ni–MgAlO catalyst was *n*-butanol, along

with some C₆–C₈ alcohols formation. In addition, ethers, esters, alkanes, and gaseous products were also detected as byproducts. The quantitative analysis results in Fig. 11 show that the ethanol conversion decreased monotonically from 26.7 to 4.6% when the WHSV was increased from 1.6 to 12.8 h⁻¹. However, the product selectivities changed differently. The selectivity for *n*-butanol showed a volcano curve as a function of WHSV. When the WHSV increased in the range 1.6–3.2 h⁻¹, the butanol selectivity increased at the expense of the hexanol selectivity. This indicates that *n*-butanol or butanal participates in the reactions to form hexanol. At higher WHSV, the *n*-butanol selectivity decreased continuously, but the acetaldehyde and butanal selectivities increased correspondingly, suggesting that the C₂ and C₄ aldehydes are involved in the *n*-butanol formation. In addition, octanol selectivity decreased with WHSV increase, largely consistent with the trend of hexanol selectivity changes. Therefore, it can be conjectured that octanol is formed mainly via reactions between C₆ and C₂ reactants.

A reaction network for ethanol conversion is proposed according to these results and publications [4,15]. As shown in Scheme 1, ethanol is first dehydrogenated at Ni sites, producing acetaldehyde

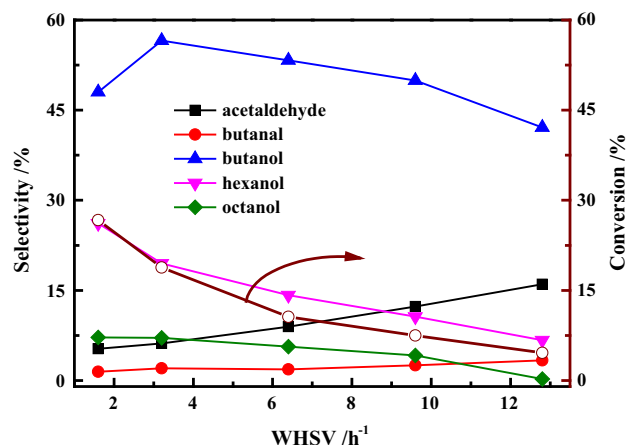
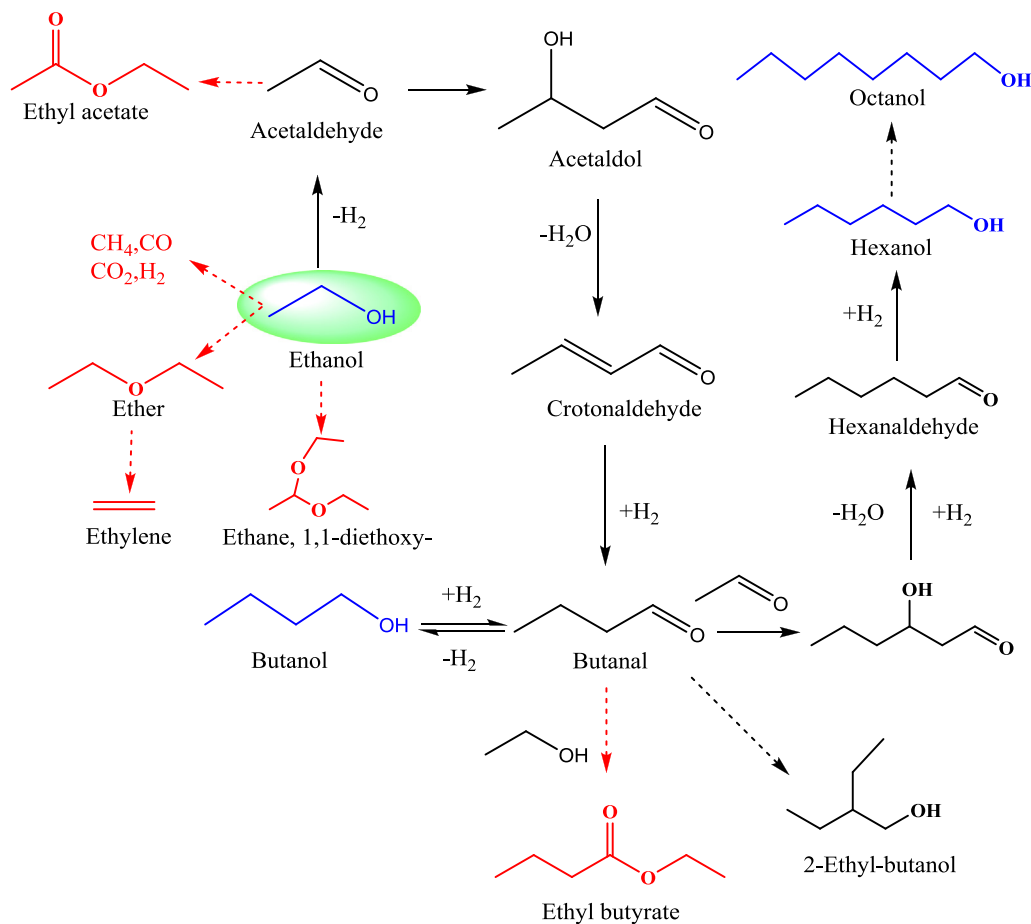


Fig. 11. Effects of WHSV on ethanol conversion over the Ni4MgAlO catalyst. Conditions: N₂ 3 MPa, 30 mL/min; 99.5% ethanol; 523 K.



Scheme 1. Proposed reaction network for ethanol conversion (red indicates main byproducts). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and hydrogen atoms that spill over onto the Ni–MgAlO catalysts. Then acetaldehyde molecules couple to acetaldo through an aldol addition reaction and are dehydrated to crotonaldehyde at the strong base sites and moderate acid sites, respectively, of Ni–MgAlO catalysts. Finally, crotonaldehyde is hydrogenated in situ to butanol by the stored hydrogen atoms. The numbers of metal, acid, and base sites should be balanced to reduce side reactions. The metallic Ni sites promote hydrogen transfer reactions. However, too many metallic Ni sites induce decomposition of ethanol and products to gaseous products; Ni is commonly used as a methanation or steam reforming catalyst. Strong acid sites induce inter/intra molecular dehydration of ethanol and produce ether and ethylene as byproducts. An inappropriate acid to base sites ratio on the catalyst causes the accumulation of aldehyde, which react with ethanol to form ester byproducts. In addition, the produced primary alcohols can react further with each other to some extent to produce long-chain alcohols.

4. Conclusions

In summary, ethanol was continuously converted to high-carbon alcohols over Ni–MgAlO catalysts in a fixed-bed reactor. Over the best catalyst, Ni₄MgAlO (19.5 wt.% Ni), the selectivities for *n*-butanol and C₄–C₈ primary alcohols were 55.2 and 85%, respectively, with 18.7% ethanol conversion and an areal reaction rate of 62.6 μmol h^{−1} m^{−2}. The catalyst was operated over a relatively wide temperature window of 473–548 K and had high stability during operation for 60 h without a reduction in butanol selectivity.

Hydrotalcite is a good precursor for preparing Ni–MgAlO catalysts with desired dispersions of Ni, acid, and base sites and gives Ni particles of size ca. 5 nm. The presence of Ni facilitated hydrogen evolution from ethanol via dehydrogenation, and also in situ hydrogenation, because of its special hydrogen adsorption and desorption properties. The composite oxides had relatively strong base and moderate acid pairs for aldol condensation reactions. The numbers of metal, acid, and base sites were well balanced on the Ni₄MgAlO catalyst, which showed the best selectivity for *n*-butanol, with enhanced ethanol conversion. Finally, a reaction network for ethanol conversion to different products is proposed.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jcat.2016.08.024>.

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